Canonical quantization (for fields) (Second quantization)

in QM dynamical variables are q, p

make them into operators with $[\hat{q}, \hat{p}] = i\hbar$

if $\mathcal{L}(q, s, t)$ Lagrangian then we define $\boxed{p = \frac{\partial \mathcal{L}}{\partial s}}$

and $\mathcal{H}(q, p, t) = s\frac{\partial \mathcal{L}}{\partial s} - \mathcal{L}$ $\quad s = s(p)$

ex: $\mathcal{L} = \frac{1}{2}ms^2 - V(q)$ $\quad p = ms \rightarrow s = \frac{p}{m}$

$$\mathcal{H} = sp - \frac{1}{2}ms^2 + V(q) = \frac{1}{2}ms^2 + V(q) = \frac{p^2}{2m} + V(q)$$

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multiple dims $\quad q = (q^1, q^2, q^3)$ $\quad [q_i, p_j] = i\delta_{ij}$

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in QFT what we do is: get field with their Lagrangian
we build associated momentum, impose commutation relations.

$\phi = \phi(\vec{x}, t)$ $\quad$ want this as equivalent of $q(t) = (q_1(t), \dots, q_n(t))$

interpret ϕ as something that for each t gives us a function of $\vec{x}$ only $\quad (t \mapsto \phi(\cdot, t))$ $\quad \phi_{\vec{x}}(t)$

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi - m^2\phi) = \frac{1}{2}(\underbrace{\dot{\phi}^2}_{(\partial_t \phi)^2} - (\vec{\nabla}\phi)^2 - m^2\phi^2)$$

think of this as a functional

$$\Pi = \frac{\delta \mathcal{L}}{\delta \dot{\phi}} = \dot{\phi} \qquad \mathcal{H} = \dot{\phi}\frac{\delta \mathcal{L}}{\delta \dot{\phi}} - \mathcal{L} = \dot{\phi}^2 - \frac{1}{2}(\dot{\phi}^2 - (\vec{\nabla}\phi)^2 - m^2\phi^2)$$

$$= \frac{1}{2}(\dot{\phi}^2 + (\nabla\phi)^2 + m^2\phi^2)$$

$$= \frac{1}{2}(\Pi^2 + (\nabla\phi)^2 + m^2\phi^2)$$

$$[\hat{\phi}(t,\vec{x}), \hat{\Pi}(t,\vec{x}')] = i\hbar\, \delta(\vec{x}-\vec{x}')$$

Want to use creation and annihilation operators (ladder operators)

Fourier expansion

ϕ satisfies the KG equation $\quad \partial_\mu \partial^\mu \phi + m^2 \phi = 0$

$$\ddot{\phi} - \nabla\cdot(\nabla\phi) + m^2\phi = 0 \quad (*)$$

$$\phi(\vec{x},t) = \mathcal{N}\int \frac{d^3\vec{k}}{(2\pi)^{3/2}} \left(A(\vec{k},t)\, e^{i\vec{k}\cdot\vec{x}} + \bar{A}(\vec{k},t)\, e^{-i\vec{k}\cdot\vec{x}} \right)$$

$\propto \varphi(k,t)$

Note: $(*)$ holds

$$\ddot{\phi}(x,t) = \mathcal{N}\int \frac{d^3\vec{k}}{(2\pi)^{3/2}} \left(\ddot{A}(\vec{k},t)\, e^{i\vec{k}\cdot\vec{x}} + \ddot{\bar{A}}(\vec{k},t)\, e^{-i\vec{k}\cdot\vec{x}} \right)$$

$$\nabla\cdot(\nabla\phi) = \mathcal{N}\int \frac{d^3\vec{k}}{(2\pi)^{3/2}} \left(A(\vec{k},t)\, \nabla\cdot\nabla e^{i\vec{k}\cdot\vec{x}} + \bar{A}(\vec{k},t)\, \nabla\cdot\nabla e^{-i\vec{k}\cdot\vec{x}} \right)$$

$$\nabla e^{i\vec{k}\cdot\vec{x}} = i\vec{k}\, e^{i\vec{k}\cdot\vec{x}} \qquad \nabla\cdot\nabla e^{i\vec{k}\cdot x} = -k^2 e^{i\vec{k}\cdot\vec{x}} \qquad (i\vec{k}\cdot i\vec{k})$$

$$\nabla\cdot(\nabla\phi) = \mathcal{N}\int \frac{d^3\vec{k}}{(2\pi)^{3/2}} \left(A(\vec{k},t)\, e^{i\vec{k}\cdot\vec{x}} + \bar{A}(\vec{k},t)\, e^{-i\vec{k}\cdot\vec{x}} \right)(-k^2)$$

$$0 = \ddot{\phi}(x,t) - \nabla\cdot\nabla\phi(x,t) + m^2\phi(x,t) = \mathcal{N}\int \frac{d^3\vec{k}}{(2\pi)^{3/2}} \underbrace{\left(\ddot{A}(k,t) + k^2 A(k,t) + m^2 A(k,t)\right)}_{=0} e^{ik\cdot x} + c.c.$$

$$\ddot{A}(\vec{k},t) + k^2 A(k,t) + m^2 A(k,t) = 0$$

$$\ddot{A} = -(m^2+k^2)A \quad \rightarrow \quad A(k,t) = A(k)\, e^{-it\omega_k} \qquad \omega_k = \sqrt{m^2+k^2}$$

$$\phi(x,t) = \mathcal{N}\int \frac{d^3\vec{k}}{(2\pi)^{3/2}} \left(A(k)\, e^{-i(t\omega_k - k\cdot x)} + \bar{A}(k)\, e^{+i(t\omega_k - k\cdot x)} \right)$$

$$\Pi(x,t) = \dot{\phi}(x,t) = \mathcal{N}\int \frac{d^3\vec{k}}{(2\pi)^{3/2}} \left(\frac{\omega_k}{i} A(k)\, e^{-i(t\omega_k - k\cdot x)} - \frac{\omega_k}{i} \bar{A}(k)\, e^{+i(t\omega_k - k\cdot x)} \right)$$

quantize: $[\hat{\varphi}(x,t), \hat{\pi}(x',t)] = i\,\delta(x-x')$ $\boxed{\hbar = 1}$

$$\hat{\phi}(x,t) = \mathcal{N}\int \frac{d^3\vec{k}}{(2\pi)^{3/2}}\left(\hat{A}(k)\, e^{-i(t\omega_k - k\cdot x)} + \hat{A}^{\dagger}(k)\, e^{+i(t\omega_k - k\cdot x)}\right)$$

$$\hat{\pi}(x,t) = \mathcal{N}\int \frac{d^3\vec{k}}{(2\pi)^{3/2}}\left(\frac{\omega_k}{i}\hat{A}(k)\, e^{-i(t\omega_k - k\cdot x)} - \frac{\omega_k}{i}\hat{A}^{\dagger}(k)\, e^{+i(t\omega_k - k\cdot x)}\right)$$

Aside: in QM. $a = q + ip$ $a^{\dagger} = q - ip$ $\rightarrow$ $[a, a^{\dagger}] = \mathbb{1}$

This is where I make the mistake $\hat{a}(k) = \sqrt{\omega_k}\,\hat{A}(k)$

$$\sqrt{\omega_k}\,\hat{\phi}(x,t) + \frac{i}{\sqrt{\omega_k}}\hat{\pi}(x,t) = 2\mathcal{N}\int \frac{d^3\vec{k}}{(2\pi)^{3/2}}\frac{1}{\sqrt{\omega_k}}\hat{a}(k)\, e^{-i(t\omega_k - k\cdot x)}$$

inverse Fourier transform $\frac{1}{2\mathcal{N}}\int \frac{d^3x}{(2\pi)^{3/2}}\left(\sqrt{\omega_k}\,\hat{\phi}(x,t) + \frac{i}{\sqrt{\omega_k}}\hat{\pi}(x,t)\right) = \hat{a}(k)$

$$\frac{1}{2\mathcal{N}}\int \frac{d^3x}{(2\pi)^{3/2}}\left(\sqrt{\omega_k}\,\hat{\phi}(x,t) - \frac{i}{\sqrt{\omega_k}}\hat{\pi}(x,t)\right) = \hat{a}^{\dagger}(k)$$

if we choose $\mathcal{N}$ appropiately $\rightarrow$ $\mathcal{N} = \frac{1}{\sqrt{2}}$

$$\boxed{[\hat{a}(k), \hat{a}^{\dagger}(k')] = \delta(k-k')}$$

vacuum state $|0\rangle$ is defined by $a(k)|0\rangle = 0$ $\forall \vec{k}$

$|\vec{k}\rangle = a^{\dagger}(k)|0\rangle$ creates a particle with momentum $\vec{k}$

$a^{\dagger}(k)\, a^{\dagger}(k')|0\rangle = |\vec{k}, \vec{k}'\rangle$

$$\langle 0|\hat{\varphi}(x,t)\,\hat{\varphi}(0,0)|0\rangle \qquad t>0$$

$$= \frac{1}{\sqrt{2}}\langle 0|\hat{\varphi}(x,t)\int\frac{d^3k}{(2\pi)^{3/2}}\frac{1}{\sqrt{\omega_k}}\left(\cancel{\hat{a}(k)}+\hat{a}^{+}(k)\right)|0\rangle$$

$$= \frac{1}{\sqrt{2}}\int\frac{d^3k}{(2\pi)^{3/2}}\frac{1}{\sqrt{\omega_k}}\langle 0|\hat{\varphi}(x,t)\,\hat{a}^{+}(k)|0\rangle$$

$$= \frac{1}{2}\int\frac{d^3k}{(2\pi)^{3/2}}\frac{d^3k'}{(2\pi)^{3/2}}\frac{1}{\sqrt{\omega_k\omega_{k'}}}\langle 0|\left(\hat{a}(k')\,e^{-i(t\omega_{k'}-k'\cdot x)}+\cancel{\hat{a}^{+}(k')}\,e^{+i(t\omega_{k'}-k'\cdot x)}\right)\hat{a}^{+}(k)|0\rangle$$

$$a|0\rangle=0 \quad\rightarrow\quad (a|0\rangle)^{+}=0 \quad\rightarrow\quad \langle 0|a^{+}=0$$

$$= \frac{1}{2}\int\frac{d^3k}{(2\pi)^{3/2}}\frac{d^3k'}{(2\pi)^{3/2}}\frac{1}{\sqrt{\omega_k\omega_{k'}}}\,e^{-i(t\omega_{k'}-k'\cdot x)}\,\langle 0|\hat{a}(k')\,\hat{a}^{+}(k)|0\rangle$$

$$\langle 0|\hat{a}(k')\,\hat{a}^{+}(k)|0\rangle - \underbrace{\langle 0|a^{+}(k)\,\hat{a}(k')|0\rangle}_{0} = \delta(k-k')\,\langle 0|0\rangle$$

$$\langle 0|\hat{\varphi}(x,t)\,\hat{\varphi}(0,0)|0\rangle = \frac{1}{2}\int\frac{d^3k}{(2\pi)^{3/2}}\frac{d^3k'}{(2\pi)^{3/2}}\frac{1}{\sqrt{\omega_k\omega_{k'}}}\,e^{-i(t\omega_{k'}-k'\cdot x)}\,\delta(k-k')$$

$$= \frac{1}{2}\int\frac{d^3k}{(2\pi)^{3/2}}\frac{1}{\omega_k}\,e^{-i(t\omega_k-k\cdot x)}$$

propagator

$$\propto\; i\lim_{\varepsilon\to 0^{+}}\int\frac{d^4p}{(2\pi)^2}\frac{e^{-ip_\mu x^\mu}}{p_0^2-(\vec{p}^{\,2}+m^2)+i\varepsilon} = i\,G(x)$$

<u>Contour integral</u>

$$\lim_{\varepsilon\to 0^{+}}\int_{-\infty}^{\infty}\frac{dp_0}{2\pi}\frac{i\,e^{-ip_0t}}{p_0^2-\omega^2+i\varepsilon} = \frac{e^{-i\omega|t|}}{2\omega}$$

$$\frac{1}{2}\int\frac{d^3k}{(2\pi)^{3/2}}\frac{1}{\omega_k}\,e^{-i(t\omega_k-k\cdot x)} = \int\frac{d^3k}{(2\pi)^{3/2}}\frac{e^{-i\omega_k t}}{2\omega_k}\,e^{ik\cdot x}$$

$$= i\lim_{\varepsilon\to 0^{+}}\int_{-\infty}^{\infty}\frac{dp_0}{2\pi}\int\frac{d^3k}{(2\pi)^{3/2}}\frac{i\,e^{-ip_0t}}{p_0^2-\omega_k^2+i\varepsilon}\,e^{ik\cdot x}$$

$\langle 0| \hat{\varphi}(x,t)\, \hat{\varphi}(0,0) |0\rangle \equiv \langle \phi(x,t)\, \phi(0,0) \rangle$ using path integral

$|k\rangle = a^{\dagger}(k)\, |0\rangle \qquad \langle k| = (a^{\dagger}(k)\, |0\rangle)^{\dagger} = \langle 0| a(k)$

$\langle 0| \hat{a}(k')\, \hat{a}^{\dagger}(k) |0\rangle \equiv \langle k' | k \rangle = \delta(k-k')$

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interpretation:

$\langle 0| \hat{\varphi}(x,t)\, \hat{\varphi}(0,0) |0\rangle = \langle 0| \hat{\varphi}(x,t) \,|\, \hat{\varphi}(0,0) |0\rangle = \langle f | i \rangle$

$|i\rangle = \hat{\varphi}(0,0)\, |0\rangle$

$|f\rangle = \hat{\varphi}(x,t)\, |0\rangle$

$\int dx\, \delta(x_1 - x)\, \delta(x_2 - x) = \delta(x_1 - x_2)$ NO sense

$\sum_k \delta_{ik}\, \delta_{jk} = \delta_{ij}$

$[a, a^{\dagger}] = \mathbb{1}$ → representation of Heisenberg group

$\Updownarrow$

$[q, p] = i$

QED: 2 groups involved.

$SO(3,1)$ → Lorentz group

$U(1) = \{ e^{i\vartheta} \mid \vartheta \in \mathbb{R} \}$

↓ gauge group

ψ Dirac field → spinor wrt $SO(3,1)$

→ it transforms as $\psi \to e^{i\vartheta} \psi$ under $U(1)$ transformations